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SUBJECT OF THE INVENTION
Hydrolysis rate detecting device and detecting method thereof

[**0001**] Technical field

[**0002**] The invention relates to the technical field of chemical detection, in particular to a hydrolysis rate detecting device and a detecting method thereof.

[**0003**] Background technique

[**0004**] The hydrolysis reaction of chemical is closely related to its persistence in the environment. It is one of the important judgments that affect the fate of organic matter in the environment, and it is also an important indicator to evaluate the residual characteristics of organic matter in water. The nucleophilic group (water or OH-) in the water reacts with the electrophilic group in the organic molecule RX, replacing a leaving group X, ie: RX H2O ROHHX. The hydrolysis reaction constant is the research index of hydrolysis kinetics. From the point of view of the change of the amount of the substance before and after the reaction, the hydrolysis of the organic matter generally conforms to the first-order kinetic reaction, and the rate of disappearance of RX-d[RX]/dt is proportional to [RX]. The change in the amount of reactants is determined as a function of time. This requires the establishment of an effective quantitative method for the determination of the concentration of the reactants. For the measurement of the substance concentration, the chemical analysis method can directly and accurately measure the concentration value at each time.

[**0005**] At present, the determination of the chemical hydrolysis rate constant requires the establishment of an accurate and reliable detection method for the reactants or products, that is, the existing test methods and standards for determining the hydrolysis rate constant are only applicable to those having sufficient A compound of analytical method for accuracy and sensitivity, suitable for compounds with sufficient water solubility. The rate of hydrolysis is calculated by monitoring the change in mass of the material from time to time. However, the accurate and reliable method for detecting the concentration of the substance takes 1 to 2 minutes from sample collection to completion of concentration detection, and takes 10 to 30 minutes for the length. Immediately in response to water, a substance with a fast hydrolysis rate and a short half-life has a large error by detecting a change in the concentration of the reactant or product.

[0006] Summary of the invention

[0007] In view of the above deficiencies in the prior art, the present invention is directed to a hydrolysis rate detecting method and apparatus for indirectly determining a substance concentration using physical properties and substance concentrations in a single value property.

[0008] In order to solve the above problems, the present invention provides a hydrolysis rate detecting device including a hydrolysis reaction generating device, a gas purifying device, a gas pressure detecting device, a first connecting pipe and a second connecting pipe, and the hydrolysis reaction generating device passes through The first connecting pipe is fixedly connected to the gas purifying device, and the gas purifying device is fixedly connected to the gas pressure detecting device through the second connecting pipe.

[0009] The hydrolysis reaction generating device includes a separator funnel and a starter generator.

[0010] The lower end of the Kaipu generator is provided with a thermostatic bath, and the lower end of the Kaipu generator is disposed in the thermostatic bath.

[0011] The starter generator includes an upper bottle body and a lower bottle body, the bottle body of the upper bottle body is provided with an exhaust port, and the bottle body of the lower bottle body is provided with a sample filling port.

[0012] A desiccant is disposed in the upper bottle body.

[0013] A sealing plug is sleeved on the neck of the separator funnel and seals the starter generator. The first connecting pipe is sleeved at one end with a sealing plug and seals the starter generator.

[0014] The gas purifying device includes a gas washing bottle and an exhaust pipe, and the exhaust pipe is disposed at an upper end of the gas washing bottle.

[0015] The gas pressure detecting device comprises a U-shaped pressure gauge and a support base, and the U-shaped pressure gauge is disposed on the support base.

[0016] a first sealing ring sleeve is disposed at a fixed connection between the first connecting pipe and the gas purifying device, and a second sealing ring sleeve is disposed at a fixed connection between the second connecting pipe and the gas purifying device and the gas pressure detecting device .

[0017] The hydrolysis rate detecting method of the hydrolysis rate detecting device comprises the following steps:

[0018] a. Air tightness inspection: water is added to the lower bottle body, the gas cylinder and the U-type pressure gauge, and the water injected into the U-tube pressure gauge makes the liquid level at both ends flush with each other, and connects the components. The water was slowly injected into the Kaipu generator from the separator funnel, and the liquid level of the U-type pressure gauge increased with the volume of the injected water. The liquid level difference between the two ends of the U-type pressure gauge increased, which proved that the hydrolysis rate detection device was airtight and reliable;

[0019] b. Sample detection: Inject a proper amount of water from the separator funnel, so that the lower end of the separator funnel is immersed under the water surface of the Kaipu generator. Accurately weigh a certain mass of the sample to be tested and add it to the Kaipu generator from the loading port. Adjust the water level at both ends of the U-type pressure gauge to be flush, close the reaction device, start timing and measure the water level difference between the two ends of the U-type pressure gauge at each time until no bubbles are generated in the gas cylinder.

[0020] The invention has the advantages that the invention is connected by the hydrolysis reaction generating device, the gas purifying device, the gas pressure detecting device, the first connecting pipe and the second connecting pipe, and utilizes the relationship between the pressure and the volume change in the physics and the concentration of the substance. The indirect measurement of the concentration of the substance, the rapid and accurate measurement of the rate of hydrolysis of the substance, the measurement error of the invention is small, and is convenient for measurement of laboratory experimental data.

[0021] DRAWINGS

[0022] Figure 1 is a schematic view of the structure of the present invention.

[0023] Detailed ways

[0024] The technical solutions in the embodiments of the present invention will be clearly and completely described in the following with reference to the accompanying drawings.

[0025] As shown in FIG. 1, a hydrolysis rate detecting device includes a hydrolysis reaction generating device 1, a gas purifying device 2, a gas pressure detecting device 3, a first connecting pipe 4, and a second connecting pipe 5, and the hydrolysis reaction generating device 1 The first connecting

pipe 4 is fixedly connected to the gas purifying device 2, and the gas purifying device 2 is fixedly connected to the gas pressure detecting device 3 via the second connecting pipe 5.

[**0026**] The hydrolysis reaction generating device 1 includes a separator funnel 11 and a starter generator 12.

[**0027**] The lower end of the starter generator 12 is provided with a thermostatic bath 6, and the lower end of the Kaipu generator 12 is disposed in the thermostatic bath 6. The function of the thermostatic bath 6 is mainly to make the internal temperature of the Kaipu generator 12 constant, thereby ensuring the Kaipu generator. 12 The internal sample to be tested has a constant reaction rate.

[**0028**] The ejector generator 12 includes an upper bottle body 13 and a lower bottle body 14. The bottle body of the upper bottle body 13 is provided with an exhaust port 15, and the bottle body of the lower bottle body 14 is provided with a sample port. 16.

[**0029**] The upper bottle body 13 is provided with a desiccant 7, and the desiccant 7 is mainly used for drying the generated gas. The desiccant 7 should not be selected to react with the gas to be tested. The desiccant 7 can be selected from silica gel or the like.

[**0030**] The separator hopper 11 is sleeved with a sealing plug and seals the ejector generator 12, and the first connecting tube 4 is sleeved at one end with a sealing plug and seals the urging generator 12, and the setting mainly serves as a sealing function.

[**0031**] The gas purifying device 2 includes a gas scrubbing bottle 21 and an exhaust pipe 22, and the exhaust pipe 22 is disposed at an upper end of the scrubbing bottle 21.

[**0032**] The gas pressure detecting device 3 includes a U-shaped pressure gauge 31 and a support base 32. The U-shaped pressure gauge 31 is disposed on the support base 32.

[**0033**] a first sealing ring sleeve 8 is disposed at a fixed connection between the first connecting pipe 4 and the gas purifying device 2, and the second connecting pipe 5 is disposed at a fixed connection with the gas purifying device 2 and the gas pressure detecting device 3 There is a second seal ring sleeve 9.

[**0034**] The hydrolysis rate detecting method of the hydrolysis rate detecting device comprises the following steps:

[**0035**] a. Airtightness inspection: water is added to the lower bottle body 14, the gas cylinder 21 and the U-type pressure gauge 31, and the water injected into the U-tube pressure gauge 31 makes the

liquid level at both ends flush and connected. For each component, water is slowly injected into the separator generator 12 from the separator funnel 11, and the liquid level of the U-type pressure gauge 31 increases with the volume of the injected water, and the liquid level difference between the two ends of the U-type pressure gauge 31 increases, demonstrating the hydrolysis rate. The detection device is airtight and reliable;

[**0036**] b. Sample detection: an appropriate amount of water is injected from the separator funnel 11 so that the lower end of the separator funnel 11 is immersed under the water surface of the Kaipu generator 12, and the sample to be tested of a certain quality is accurately weighed and added from the sample inlet 16 The generator 12 is adjusted so that the water surface at both ends of the U-type pressure gauge 31 is flush, the reaction device is closed, the timing is started, and the water level difference between the two ends of the U-type pressure gauge 31 is measured until the gas bottle 21 is no longer present. Bubbles are generated.

[**0037**] Specifically, the sample to be tested generates a gas through the hydrolysis reaction generating device in the apparatus of the present invention, and the gas passes through a desiccant 7 (such as silica gel) to remove part of the moisture, and then enters the first connecting pipe 4 through the exhaust port, and is transferred to the gas purification. In the device 2, the gas is purified by the gas purifying device 2, and finally the pure gas is introduced into the U-type pressure gauge 31 through the exhaust pipe, which causes the originally balanced water column in the U-type pressure gauge 31 to change, and we can calculate by the change value. Reaction rate constant and gas production amount.

[**0038**] The reaction rate constant is calculated: the amount of water in the hydrolysis process is much larger than the sample amount, and the hydrolysis reaction can be treated according to the first-order reaction, and the hydrolysis rate is in accordance with the following relationship: lnCt=-Kt +lnC0, to generate a simple gas by hydrolysis. To indicate, there is lnCt=Kt+lnC0, and the data obtained by converting
[image]
according to PV=nRT is plotted as
[image]
for t, and the rate constant K is calculated by slope (S): K=2.303S, hydrolysis half-life (T1/2) is: T1/2=2.303lg2/K=0.693/K (t—reaction time min|); P, Pt—pressure (Pa) generated by CO2 at a constant pressure and at time t.

[**0039**] Calculation of gas production amount: According to PV=nRT, when the amount of production changes, the pressure P also changes. Under the same volume v, thermodynamic constant R, and ambient temperature T, $\Delta n=\Delta P\cdot V$ is obtained. /(RT), the ΔP value is obtained by the pressure test

system, and the temperature test system obtains the value of the temperature T, that is, the value of the gas generation amount Δn can be calculated.

1.A hydrolysis rate detecting device comprises a hydrolysis reaction generating device (1), a gas purifying device (2), a gas pressure detecting device (3), a first connecting pipe (4) and a second connecting pipe (5), characterized in that The hydrolysis reaction generating device (1) is fixedly connected to the gas purifying device (2) through the first connecting pipe (4), and the gas purifying device (2) passes through the second connecting pipe (5) It is fixedly connected to the gas pressure detecting device (3).

2.The hydrolysis rate detecting apparatus according to claim 1, wherein said hydrolysis reaction generating means (1) comprises a separator funnel (11) and a starter generator (12).

3.The hydrolysis rate detecting device according to claim 2, wherein the lower end of the starter generator (12) is provided with a constant temperature bath (6), and the lower end of the starter generator (12) is disposed in the constant temperature bath (6) .

4.A hydrolysis rate detecting device according to claim 3, wherein said ejector generator (12) comprises an upper bottle body (13) and a lower bottle body (14), and said upper bottle body (13) bottle An exhaust port (15) is disposed on the body, and a filling port (16) is disposed on the bottle body of the lower bottle body (14).

5.The hydrolysis rate detecting device according to claim 4, wherein a desiccant (7) is disposed in said upper bottle body (13).

6.The hydrolysis rate detecting device according to claim 2, wherein the separator funnel (11) has a sealing plug on the neck and seals the starter generator (12), and the first connecting pipe (4) has one end. The sealing plug is sealed and the starter generator (12) is sealed.

7.The hydrolysis rate detecting device according to claim 1, wherein said gas purifying device (2) comprises a gas washing bottle (21) and an exhaust pipe (22), and said exhaust pipe (22) is provided for washing The upper end of the gas cylinder (21).

8.A hydrolysis rate detecting device according to claim 1, wherein said gas pressure detecting means (3) comprises a U-type pressure gauge (31) and a support base (32), said U-type pressure gauge (31) being set On the support base (32).

9.The hydrolysis rate detecting device according to claim 1, wherein the first connecting pipe (4) and the gas purifying device (2) are fixedly connected with a first sealing ring sleeve (8), A second seal ring (9) is disposed at a fixed connection between the second connecting pipe (5) and the gas purifying device (2) and the gas pressure detecting device (3).

10.A method for detecting a hydrolysis rate of a hydrolysis rate detecting device according to any one of claims 1-9, comprising the steps of:

a. Airtightness check: water is added to the lower bottle body (14), the gas cylinder (21) and the U-type pressure gauge (31), and the water injected into the U-tube pressure gauge (31) makes two The liquid level of the end is flush, and the components are connected. The water is slowly injected into the starter generator (12) from the separator funnel (11), and the volume of the U-type pressure gauge (31) increases with the volume of the injected water. U-shaped The liquid level difference between the two ends of the pressure gauge (31) increases, which proves that the hydrolysis rate detecting device is airtight and reliable;

b. Sample detection: Inject a proper amount of water from the separator funnel (11), so that the lower end of the separator funnel (11) is immersed under the water surface of the Kaipu generator (12) to accurately weigh a certain mass of the sample to be tested. The addition port (16) is added to the Kaipu generator (12), adjusted so that the water surface at both ends of the U-type pressure gauge (31) is flush, the reaction device is closed, the timing is started, and the ends of the U-type pressure gauge (31) are measured at each time. The water level is poor until no bubbles are formed in the scrubber (21).